

Literature status of the amenable operator algebra similarity question

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Lane 4, `agent_lane_04`, model GPT5.6

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Status. Literature already answered. This is a retrieval and identification note, not a new proof or counterexample.

Source question

Choi's paper [1], arXiv:1012.4259v3, asks in its abstract and Introduction on PDF page 1 whether every closed amenable subalgebra of $B(H)$ must be similar to an amenable C^* -algebra. It emphasizes that the question was then open even for singly-generated algebras. On PDF page 2 the paper also observes that extending its finite-von-Neumann-algebra theorem to the semifinite setting would settle the commutative case.

Identification

The unrestricted question has a negative answer. Choi, Farah, and Ozawa [2], Theorem 1 on PDF page 2, construct a unital amenable operator algebra $\mathcal{A}$ that is not isomorphic to any C^* -algebra. If SAS^{-1} were a C^* -algebra for some invertible S , conjugation by S would be an algebra isomorphism from $\mathcal{A}$ onto a C^* -algebra. Thus their example is not similar to a C^* -algebra.

The abelian and singly-generated cases have a positive answer. Marcoux and Popov [3], Theorem 2.10 on PDF page 11, prove that every closed abelian amenable operator algebra is similar to a C^* -algebra. Their Corollary 2.11 on the same page states that, for $T \in B(H)$, the norm-closed unital algebra $\mathcal{A}_T$ generated by T is amenable if and only if T is similar to a normal operator whose spectrum is a Lavrentieff set.

The provenance match is explicit rather than inferred: Marcoux and Popov cite Choi's source result and state that the unrestricted question was resolved negatively by Choi–Farah–Ozawa.

Scope limitation

The negative example is nonseparable and nonabelian. Both supporting papers state that the separable nonabelian version remains open. A bounded arXiv search through July 2026 found no later resolution, but this compact packet does not claim an exhaustive survey of that narrower problem.

References

- [1] Y. Choi, *On commutative, operator amenable subalgebras of finite von Neumann algebras*, arXiv:1012.4259v3 (2011).
- [2] Y. Choi, I. Farah, and N. Ozawa, *A nonseparable amenable operator algebra which is not isomorphic to a C^* -algebra*, arXiv:1309.2145 (2013).
- [3] L. W. Marcoux and A. I. Popov, *Abelian, amenable operator algebras are similar to C^* -algebras*, arXiv:1311.2982v2 (2013).